STAGE 4 – DOCUMENTATION, INSIGHTS AND PRESENTATION

Labour Market Analytics: Employment Resilience Across Indian States

1. Dashboard

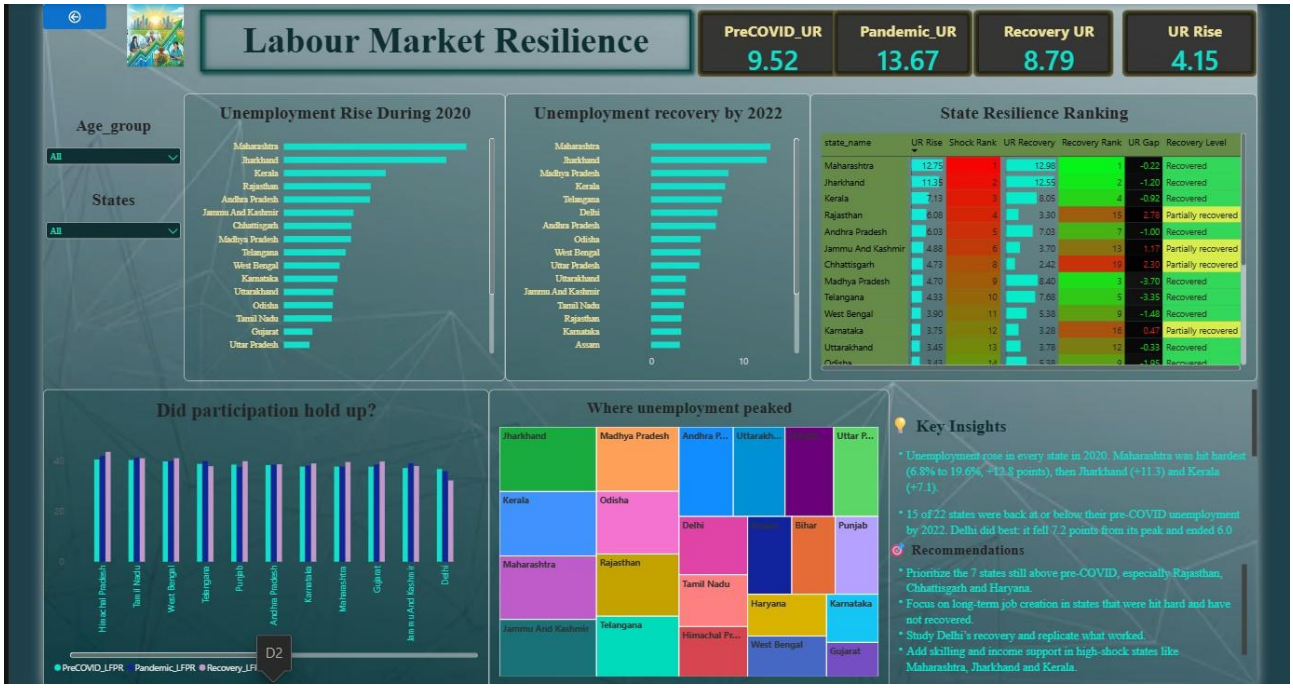
The final Power BI dashboard has two pages. Page 1 gives the overall labour market picture; Page 2 explains how each state was hit by COVID-19 and how far it recovered. Both pages share the same filters (Age Group, States, and Gender on Page 1), so every chart on a page changes together.

Python and Power BI workflow:

- **Python (Google Colab):** merged the four PLFS files, cleaned the data, treated outliers and skewness, created the phase column (Pre-COVID 2019, Pandemic 2020, Recovery 2022) and carried out the exploratory analysis (histograms, boxplots, correlation heatmap, pair plot, state-by-year heatmap).
- **Cleaned output:** Cleaned_labour_market_data.csv (4,347 records × 17 columns) was loaded into Power BI as the fact table.
- **Power BI:** DAX measures (phase averages, UR Rise, UR Recovery, UR Gap, Shock Rank, Recovery Rank, Recovery Ratio, Recovery Level) drive the KPI cards, charts and the State Resilience Ranking table.



Page 1 – Labour Market Overview



Page 2 – Labour Market Resilience

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2. Summary :

- **COVID-19 caused a sharp but temporary job shock.** Average unemployment rose from about 9.5% before the pandemic to about 13.7% in 2020, then fell to about 8.8% by 2022, slightly better than where it started.
- **Every state was hit, but not equally.** Maharashtra, Jharkhand and Kerala saw the biggest increases. A few states, such as Himachal Pradesh, saw a delayed rise instead of a 2020 spike.
- **Most states recovered, some did not.** 15 of 22 states were back at or below their pre-COVID unemployment by 2022. Seven were still above it, led by Rajasthan, Chhattisgarh and Haryana.
- **Women and men experience the labour market very differently.** About 17 in 100 women take part in the labour force, compared with about 57 in 100 men, and women's unemployment is higher.
- **Participation and employment move together.** When more people join the labour force, more people are also working (correlation 0.99). Farm-heavy states tend to do worse than industry-heavy states.
- **The pandemic hit jobs more than participation.** Participation barely changed, while unemployment moved sharply, so the shock was about losing work, not about people leaving the labour force.

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3. Key Insights

Type	Insight	What it means
Trend	Unemployment rose from 9.52% (pre-COVID) to 13.67% (pandemic) and fell to 8.79% (recovery), a rise of 4.15 points.	The shock was large but reversed, so the labour market was resilient overall.
Trend	Nationally, LFPR and WPR kept rising through 2024 (LFPR from 36.0% in 2019 to 39.5% in 2024 Q1).	Participation and employment are growing steadily after the shock.

Anomaly	Maharashtra rose from 6.8% to 19.6% (+12.8 points), the largest shock, then Jharkhand (+11.3) and Kerala (+7.1).	Resilience varies a lot by state, so national averages hide local problems.
Anomaly	Rajasthan (+2.8) and Chhattisgarh (+2.3) were hit hard and were still above pre-COVID levels in 2022. Himachal Pradesh had no 2020 spike but a later peak.	These states need targeted support; one recovery pattern does not fit all.
Correlation	LFPR and WPR have a correlation of 0.99. Agriculture share is negative with LFPR (-0.41) and WPR (-0.43); secondary-sector share is positive (0.56 and 0.58).	States with more industry have better participation and employment than farm-heavy states.
Gap	Men's LFPR is 57.1% against 16.9% for women; women's unemployment is 11.1% against 8.6% for men.	The gender gap is the largest structural gap in the data and it barely changed during the pandemic.

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4. Recommendations for Decision-Making

Overview page:

1. **Boost women's participation** – improve job access and childcare support (LFPR 16.9% against 57.1% for men).
2. **Tackle women's higher unemployment** – use skilling and placement programmes (11.1% against 8.6%).
3. **Create local jobs in low-participation states** such as Bihar, using Himachal Pradesh as the benchmark.
4. **Expand non-farm jobs**, since farm-heavy states show weaker participation and employment.
5. **Track LFPR and WPR with unemployment** to confirm the recovery lasts.

Resilience page:

1. **Prioritize the seven states still above pre-COVID**, especially Rajasthan, Chhattisgarh and Haryana.
2. **Focus on long-term job creation** in states that were hit hard and have not recovered.
3. **Study Delhi's recovery** (down 7.2 points from its peak, ending 6.0 below pre-COVID) and replicate what worked.
4. **Add skilling and income support** in high-shock states such as Maharashtra, Jharkhand and Kerala.
5. **Track participation alongside unemployment** so that falling unemployment does not hide people leaving the labour force.

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5. The Final Story

Starting point. Before COVID-19, the average unemployment rate was 9.52%. Participation was steady, with a large gap between men and women. The Overview page shows this baseline through its KPI cards, trend lines and the male versus female chart.

The shock. In 2020 unemployment jumped to 13.67%, a 4.15-point rise. The Resilience page shows that every state was hit, with Maharashtra, Jharkhand and Kerala at the top of the "Unemployment Rise During 2020" chart, and the treemap shows where unemployment peaked.

The recovery. By 2022 average unemployment had fallen to 8.79%, below the pre-COVID level. The State Resilience Ranking table shows who recovered: Shock Rank, Recovery Rank and UR Gap tell how hard a state was hit and where it ended up, and the Recovery Level colours (Recovered, Partially recovered, Slowly

recovered, Not recovered) make the winners and laggards visible at a glance. 15 of 22 states are back at or below their starting point.

What is left to fix. Seven states are still above pre-COVID levels, and the gender gap in participation (57.1% for men against 16.9% for women) and the weaker outcomes in farm-heavy states remain. Participation did not collapse during the pandemic, so the problem is the availability of jobs, not people leaving the workforce.

The decision. Policy should focus on three things: women's participation, non-farm jobs, and targeted support for the states that have not yet recovered, while monitoring participation alongside unemployment.

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6. Conclusion

The dashboard turns four PLFS datasets into one story: COVID-19 caused a sharp but temporary rise in unemployment, most states recovered by 2022, and the gender gap and state-level differences remain.

The interactive filters let decision-makers explore these patterns by state, gender and age group, and the recommendations point to where action is most needed.